Year 9 Mathematics

Unit 3: Sequences and Algebra

3.2 Geometric Sequences

Geometric Sequences

STARTER · 5 MIN

Date: _____ **Title: Geometric Sequences**

LEARNING OBJECTIVES

- Recognise and continue geometric sequences
- Use fraction, decimal and negative ratios
- Solve problems with geometric sequences

STARTER *Answer in your book. No calculators.*

1. Work out $3 \times 4 \times 4$
2. Work out $54 \div 3$
3. Work out 0.6×0.5
4. Work out -2×3

Starter — Answers

STARTER

Tick or correct in green pen

1. **48**
2. **18**
3. **0.3**
4. **−6**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can recognise and continue geometric sequences	☐
SILVER	I can use fraction, decimal and negative ratios	☐
GOLD	I can solve problems with geometric sequences	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Geometric Sequences

INTRODUCTION · 3 MIN

In a geometric sequence you multiply by the same number each time.

Common ratio

The number you multiply by.

2, 6, 18, 54, … (× 3)

Find the ratio

Divide a term by the term before it.

18 ÷ 6 = 3

Fraction ratio

The terms get smaller.

64, 32, 16, … $(\times \frac{1}{2})$

Negative ratio

The signs swap each time.

1, −2, 4, −8, … (× −2)

Examples 1 and 2 — Continuing Geometric Sequences

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint common ratio = term ÷ term before

Example 1

2, 6, 18, 54, …

(a) Find the common ratio.

(b) Write the next two terms.

Example 2

The first term is 5 and the common ratio is 2.

(a) Write the first five terms.

(b) How many terms are below 100?

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Divide a term by the one before

(a) 6 ÷ 2 = 3: ratio 3

(b) 54 × 3 = 162

162 × 3 = 486

Tip: check: 18 ÷ 6 = 3 too.

Example 2

Keep multiplying by 2

(a) 5, 10, 20, 40, 80

(b) next: 160

5 terms are below 100

Tip: list terms until you pass 100.

Task 1

PRACTICE · 9 MIN

BRONZE

Questions 1–3: the ratio and the next two terms. 4–6: as asked.

Hint common ratio = term ÷ term before

1 3, 6, 12, 24, …

2 1, 5, 25, 125, …

3 4, 12, 36, …

4 First four terms: first term 2, ratio 4

5 First four terms: first term 3, ratio 10

★ First term 3, ratio 3: how many terms below 200?

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. × 2: 48, 96
2. × 5: 625, 3125
3. × 3: 108, 324
4. 2, 8, 32, 128
5. 3, 30, 300, 3000

★ 3, 9, 27, 81, (243): 4 terms

Examples 3 and 4 — Fraction and Negative Ratios

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint the ratio can be a fraction negative ratio: signs swap

Example 3

64, 32, 16, 8, …

Find the common ratio and the next two terms.

Example 4

2, −6, 18, −54, …

Find the common ratio and the next two terms.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Divide: 32 ÷ 64

ratio = $\frac{1}{2}$ (halve each time)

$8 \times \frac{1}{2} = 4$

$4 \times \frac{1}{2} = 2$

Tip: halving is the same as ÷ 2

Example 4

Divide: −6 ÷ 2

ratio = −3

−54 × −3 = 162

162 × −3 = −486

Tip: a negative ratio swaps the sign.

Task 2

PRACTICE · 9 MIN

SILVER

Find the common ratio and the next two terms

Hint the ratio can be a fraction
negative ratio: signs swap

1. 80, 40, 20, …
2. 1000, 100, 10, …
3. 0.6, 0.3, 0.15, …
4. 1, −2, 4, −8, …
5. 3, −15, 75, …

★ 243, 81, 27, …

Task 2 — Answers

CORRECTIONS

SILVER *Tick or correct in green pen*

1. $\times \frac{1}{2}$: 10, 5

2. × 0.1: 1, 0.1

3. × 0.5: 0.075, 0.0375

4. × −2: 16, −32

5. × −5: −375, 1875

★ $\times \frac{1}{3}$: 9, 3

Examples 5 and 6 — Types of Sequence and Patterns

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint arithmetic: + same geometric: × same
quadratic: 2nd differences equal

Example 5

(3 marks)

Is each sequence arithmetic, geometric or quadratic?

(a) 3, 7, 11, 15, …

(b) 3, 6, 12, 24, …

(c) 1, 4, 9, 16, …

Example 6

(2 marks)

Pattern 1 has 1 cross, pattern 2 has 3 crosses and pattern 3 has 9 crosses.
Waleed says pattern 5 has 243 crosses. Is he correct?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

+ same: arithmetic, × same: geometric

(a) + 4 each time: arithmetic

(b) × 2 each time: geometric

(c) 2nd differences 2: quadratic

Tip: test + first, then ×.

Example 6

Geometric: × 3 each time

1, 3, 9, 27, 81

pattern 5 has 81 crosses

No: 243 is pattern 6

Tip: list the terms to check.

Task 3

PRACTICE · 9 MIN

GOLD

Questions 1–4: arithmetic, geometric or quadratic? 5–6: patterns

1 5, 10, 20, 40, …

2 5, 10, 15, 20, …

3 2, 6, 18, 54, …

4 2, 5, 10, 17, …

5 Patterns have 2, 4, 8 tiles.
How many tiles in pattern 6?

★ Patterns have 1, 4, 16 dots.
Is pattern 5 256 dots? Explain.

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **geometric (× 2)**
2. **arithmetic (+ 5)**
3. **geometric (× 3)**
4. **quadratic**
5. **2, 4, 8, 16, 32, 64: 64 tiles**

★ **Yes: 1, 4, 16, 64, 256**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the next term: 4, 20, 100, …

SILVER Find the common ratio: 81, 27, 9, …

GOLD Is 7, 14, 28, 56, … arithmetic or geometric?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **500**

SILVER $\times \frac{1}{3}$

GOLD **geometric (× 2)**

Exam Practice — Geometric Sequences

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint common ratio = term ÷ term before

Question 1

(2 marks)

Here is a geometric sequence.
5, 15, 45, 135, …

(a) Write down the common ratio. (1)

.....

(b) Write down the next term. (1)

.....

Question 2

(1 mark)

The first term of a geometric sequence is 2. The common ratio is 4. How many terms in the sequence are below 50?

.....

Exam Practice — Fraction and Negative Ratios

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint the ratio can be a fraction
negative ratio: signs swap

Question 3

(2 marks)

Here is a geometric sequence.

96, 48, 24, 12, …

(a) Write down the common ratio. (1)

.....

(b) Write down the next two terms. (1)

.....

Question 4

(2 marks)

Here is a geometric sequence.

−1, 4, −16, 64, …

Find the next term.

.....

Exam Practice — Patterns

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint arithmetic: + same geometric: × same
quadratic: 2nd differences equal

Question 5

(2 marks)

Pattern 1 has 2 crosses, pattern 2 has 8 and pattern 3 has 32.
Waleed says pattern 5 has 2048.

(a) Is it geometric? Give a reason. (1)

.....

(b) Explain why Waleed is wrong. (1)

.....

Question 6

(3 marks)

Is each sequence arithmetic, geometric or quadratic?

(a) 4, 12, 36, 108, … (1)

.....

(b) 4, 9, 14, 19, … (1)

.....

(c) 2, 5, 10, 17, … (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 3 B1

(b) 405 B1

Tip: 15 ÷ 5 = 3

Question 2

2, 8, 32, (128): 3 terms B1

Tip: list the terms until you pass 50.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $\frac{1}{2}$ — B1

(b) 6, 3 — B1

Tip: 48 ÷ 96 = 0.5, so halve each term

Question 4

ratio −4 — M1

64 × −4 = −256 — A1

Tip: the sign swaps each time.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) Yes: × 4 each time B1

(b) 2, 8, 32, 128, 512

pattern 5 has 512 B1

Tip: 2048 is pattern 6.

Question 6

(a) geometric B1

(b) arithmetic B1

(c) quadratic B1

Tip: (c) differences 3, 5, 7